

to absorption, diffraction, scattering, refraction, beam wander, scintillation, beam dispersion, etc that may be caused by physical composition of atmosphere, changes in refractive indices, building sway, fog, rain, dust, smog, snow, interference from background light sources (including the sun), shadowing, etc (Fig. 4).

1. Environmental factors.

Fog. Fog is composed of water droplets between a few and a hundred microns in diameter, which allow absorption, scattering and reflection to occur when laser beams pass through it. Fog is a major challenge to FSO communications and can result in atmospheric attenuation of signal (10-100 dB/km). Heavy fog can result in a complete outage of the service.

Low clouds, rain, snow and dust. Low clouds are very similar to fog and may accompany rain and snow that may degrade optical signal significantly.

Rain. Rain can also degrade the performance of FSO transmission. Raindrop sizes are larger than fog so may cause scattering of optical signal. Extremely heavy rain can entirely disrupt the link. Water sheeting on windows can also deteriorate signal.

Heavy snow. Heavy snow may cause whiteout conditions and may result in ice build-up on windows. Laser beams are affected by heavy snow and rain that can cause attenuation to the laser beams of 60-1000 dB/km.

Sandstorms. Sand/dust particles can alter signal path to a finite extent. In desert areas, sandstorms occur very often but they occur rarely in urban areas.

The phenomenon that causes signal degradation due to atmospheric conditions is explained in the following lines:

Absorption. Absorption occurs when suspended water molecules in the terrestrial atmosphere extinguish photons. This causes a decrease in the power density of the FSO beam and directly affects the availability of telecom services. However, typical wavelength values of FSO systems are 785nm, 850nm and 1500nm. These wavelengths correspond to atmospheric

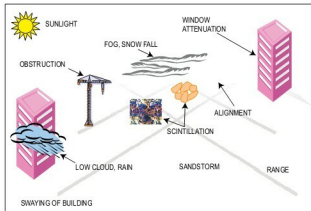


Fig. 4: Factors affecting FSO performance

windows where the attenuation is very small; thus the absorption attenuation is also very small.

Scattering. Scattering is caused when a light wave collides with the scatterer. The physical size of the scatterer determines the type of scattering. When the scatterer is smaller than the wavelength of light, it is known as Rayleigh scattering, which happens due to atmospheric gas molecules. When the scatterer is of comparable size to the signal wavelength, Mie scattering occurs, which is particularly caused by aerosol particles.

Scintillation. Heated air rising from the earth or man-made devices such as heating ducts creates temperature variations amongst different air pockets. Propagation through air pockets of varying temperature, density and refraction index causes beam spreading and wandering. Scintillation is temporal and can cause spatial variation in light intensity, which leads to increased error rate but not complete service outage.

2. Physical obstructions. Flying birds or some temporary obstruction in the path of FSO signal can temporarily block a single optical beam, but this tends to cause only short interruptions, and transmissions are easily and automatically resumed.

3. Building sway/seismic activity. The movement/swaying of tall structures/buildings due to winds, coal mine blasts and seismic activity can upset receiver and transmitter alignment. Usually building sway is

not more than 4-10 milliradians, but such a small misalignment can also increase bit error rate. In such situations, beam divergence is kept larger than expected building motion. Automatic pointing and tracking is the best option for exploiting optimum performance of an FSO system.

4. Window attenuation. In some cases, where high-altitude rooftop weather losses are more, FSO systems are installed in windows. But, it has been observed that uncoated glass attenuates almost four per cent of signal per surface due to reflection. Tinted or insulated windows can have much greater attenuation. So, a trade-off between these two options (window or rooftop installation) should be achieved.

To sum up

During the last few years, FSO technology has become one of the hottest topics in the telecommunication industry because it has the most promising capabilities to the last-mile bottleneck problems. Technology and economics both favour the optical wireless technology. Although there are several factors that degrade FSO signal performance as well. But, with the advancement in the FSO technology, fourth-generation state-of-the-art FSO systems employing multibeam, multipath architecture, laser auto tracking, devices with large fade margin (extra power reserved for fog, rain, smog, etc), network with shorter link distances, use of eye-safe lasers with limited laser power density, etc have addressed issues encountered by first-generation FSO systems. ●

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